Experiment No.5

Interactive Shell Script for Simple Calculator

Aim: To Write and execute an interactive shell script that performs basic arithmetic operations as a simple calculator.

Theory:

An interactive shell script is a script that accepts input from the user during execution. The read command is used to take input from the keyboard, making the script user-friendly and interactive.

A simple calculator performs basic arithmetic operations such as addition, subtraction, multiplication, and division using user-provided numbers.

1. The read Command

The read command stores user input into variables.

Syntax:

Bash

Read variable_ name

Example:

Bash

Echo “Enter a number:”

Read num

2. Arithmetic Expansion

Arithmetic calculations in Bash are performed using:

Bash

$((expression))

Example:

Bash

Result=$((a+b))

3. Case Statement

The case statement is used to execute one of several blocks of commands based on the user's choice. It is more readable than multiple if-else statements when there are many options.

Syntax:

Bash

case variable in

1)

commands

;;

2)

commands

;;

*)

commands

;;

esac

4. Arithmetic Operators

Operator	Meaning
+	Addition
-	Subtraction
*	Multiplication
/	Division
%	Modulus (Remainder)

Viva Questions:

1. What isthe use of the case statement?
2. What are arithmetic operators in Bash?
3. How is userinput stored in a shell script?
4. How do you display output in a shell script?
5. What is the purpose of the echo command?
6. What is the useof ;; in a case statement?
7. How do you run a shell script from the terminal?

8. What are the advantages of interactive shell scripts?